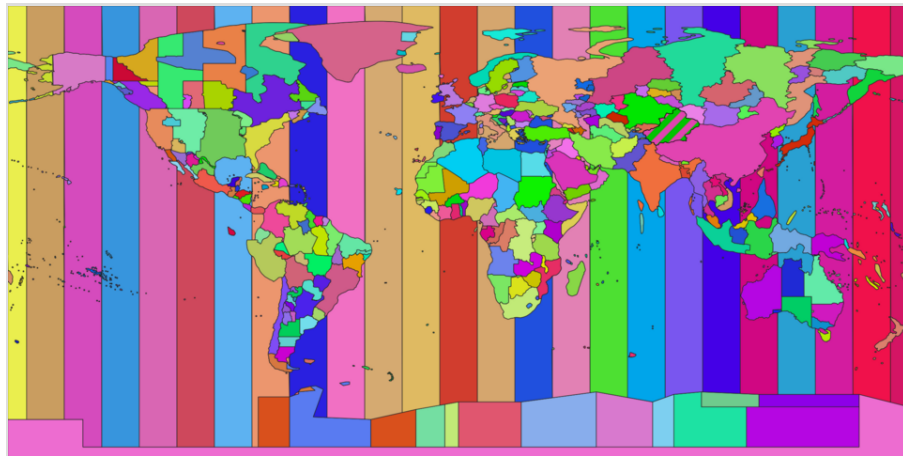


tz database

The **tz database** is a collaborative compilation of information about the world's time zones and rules for observing daylight saving time, primarily intended for use with computer programs and operating systems.^[2] Paul Eggert has been its editor and maintainer since 2005,^[3] with the organizational backing of ICANN.^[4] The tz database is also known as **tzdata**, the **zoneinfo database** or the **IANA time zone database** (after the Internet Assigned Numbers Authority), and occasionally as the **Olson database**, referring to the founding contributor, Arthur David Olson.^[5]



The tz database partitions the world into regions where local clocks all show the same time. This map was made by combining version 2023d of the List of tz database time zones with OpenStreetMap data, using open-source software.^[1]

Its uniform naming convention for entries in the database, such as *America/New_York* and *Europe/Paris*, was designed by Paul Eggert.^[6] The database attempts to record historical time zones and all civil changes since 1970, the Unix time epoch.^[7] It also records leap seconds.^[8]

The database, as well as some reference source code, is in the public domain.^[9] New editions of the database and code are published as changes warrant, usually several times per year.^[10]

Data structure

Definition of a timezone

Within the tz database, a *timezone* is any national region where local clocks have all agreed since 1970.^[11] This definition concerns itself first with geographic areas which have had consistent local clocks. A timezone is different from a region with a particular standard time offset from UTC, which is often referred to as a "time zone". Therefore, each of the timezones defined by the tz database may use multiple offsets from UTC, such as offsets for standard time and daylight saving time.^[12]

File formats

The tz database is published as a set of text files which list the rules and zone transitions in a human-readable format. For use, these text files are compiled into a set of platform-independent binary files—one per timezone. The reference source code includes such a compiler called *zic* (zone information compiler), as well as code to read those files and use them in standard APIs such as `localtime()` and `mktime()`.

Timezones

Each timezone has one or more "zone lines" in one of the tz database text files. The first zone line for a timezone gives the name of the timezone; any subsequent zone lines for that timezone leave the name blank, indicating that they apply to the same zone as the previous line. Each zone line for a zone specifies, for a range of date and time, the offset to UTC for standard time, the name of the set of rules that govern daylight saving time (or a hyphen if standard time always applies), the format for time zone abbreviations, and, for all but the last zone line, the date and time at which the range of date and time governed by that line ends.

Daylight saving time (DST) rules

The rules for daylight saving time are specified in named rule sets. Each rule set has one or more rule lines in the text files. A rule line contains the name of the rule set to which it belongs, the first year in which the rule applies, the last year in which the rule applies (or "only" if it applies only in one year or "max" if it is the rule then in effect), the type of year to which the rule applies ("- " if it applies to all years in the specified range, which is almost always the case, otherwise a name used as an argument to a script that indicates whether the year is of the specified type), the month in which the rule takes effect, the day on which the rule takes effect (which could either be a specific day or a specification such as "the last Sunday of the month"), the time of day at which the rule takes effect, the amount of time to add to the offset to UTC when the rule is in effect, and the letter or letters to use in the time zone abbreviation (for example, "S" if the rule governs standard time and "D" if it governs daylight saving time).

Names of timezones

The timezones have unique names in the form "*Area/Location*", e.g. "America/New_York". A choice was also made to use English names or equivalents, and to omit punctuation and common suffixes. The underscore character is used in place of spaces. Hyphens are used where they appear in the name of a location. The *Area* and *Location* names each have a maximum length of 14 characters.^{[13][14]}

Area

Area is the name of a continent, an ocean, or "Etc". The continents and oceans used are Africa, America, Antarctica, Arctic, Asia, Atlantic, Australia, Europe, Indian, and Pacific.

The oceans are included since some islands are hard to connect to a certain continent. Some are

geographically connected to one continent and politically to another. See also [Boundaries between continents](#).

The special area of "Etc" is used for some administrative zones, particularly for "Etc/UTC" which represents [Coordinated Universal Time](#). In order to conform with the [POSIX](#) style, those zone names beginning with "Etc/GMT" have their sign reversed from the standard [ISO 8601](#) convention. In the "Etc" area, zones west of GMT have a positive sign and those east have a negative sign in their name (e.g "Etc/GMT-14" is 14 hours ahead of GMT).

Location

Location is the name of a specific location within the area – usually a city or small island.

Country names are not normally used in this scheme, primarily because they would not be robust, owing to frequent political and boundary changes. The names of large cities tend to be more permanent.^[15] Usually the most populous city in a region is chosen to represent the entire timezone, although another city may be selected if it is more widely known, and another location, including a location other than a city, may be used if it results in a less ambiguous name.^[16] In the event that the name of the location used to represent the timezone changes, the convention is to create an alias^[17] in future editions so that both the old and new names refer to the same database entry.

In some cases the *Location* is itself represented as a compound name, for example the timezone "America/Indiana/Indianapolis". Three-level names include those under "America/Argentina/...", "America/Kentucky/...", "America/Indiana/...", and "America/North_Dakota/...".

The location selected is representative of the entire area; that is, the current time at the location is the same as the current time in the entire zone. However, this does not necessarily hold for periods before 1970. That is, the time zone rules are only guaranteed to be correct for the *named location* for times before 1970; if there were time differences within the area before 1970, the time zone rules only apply in the named location for that period.

Examples

Name	Explanation
America/Costa_Rica	name of country used because the name of the largest city (and capital city) San José is ambiguous
America/New_York	Space replaced with underscore
Asia/Kolkata	name of city of Kolkata used, because it was the most populous city in the zone at the time the zone was set up ^[18]
Asia/Sakhalin	name of island used, because largest city, Yuzhno-Sakhalinsk , has more than 14 characters
America/Bahia_Banderas	"de" removed from Bahia de Banderas , because correct name has more than 14 characters
Antarctica/DumontDUrville	the apostrophe is removed. The space would normally be replaced with "_", but the name would then exceed 14 characters.

Example zone and rule lines

These are rule lines for the standard United States daylight saving time rules, rule lines for the daylight saving time rules in effect in the US Eastern Time Zone (called "NYC" as New York City is the city representing that zone) in some years, and zone lines for the America/New_York timezone, as of release version *tzdata2011n* of the time zone database. The zone and rule lines reflect the history of DST in the United States.

```

# Rule  NAME  FROM  TO    TYPE  IN    ON      AT      SAVE  LETTER/S
Rule    US    1918  1919  -      Mar   lastSun 2:00    1:00   D
Rule    US    1918  1919  -      Oct   lastSun 2:00    0       S
Rule    US    1942  only  -      Feb    9       2:00    1:00   W # War
Rule    US    1945  only  -      Aug    14      23:00u  1:00   P # Peace
Rule    US    1945  only  -      Sep    30      2:00    0       S
Rule    US    1967  2006  -      Oct   lastSun 2:00    0       S
Rule    US    1967  1973  -      Apr   lastSun 2:00    1:00   D
Rule    US    1974  only  -      Jan    6       2:00    1:00   D
Rule    US    1975  only  -      Feb    23      2:00    1:00   D
Rule    US    1976  1986  -      Apr   lastSun 2:00    1:00   D
Rule    US    1987  2006  -      Apr   Sun>=1 2:00    1:00   D
Rule    US    2007  max   -      Mar   Sun>=8 2:00    1:00   D
Rule    US    2007  max   -      Nov   Sun>=1 2:00    0       S
....
# Rule  NAME  FROM  TO    TYPE  IN    ON      AT      SAVE  LETTER
Rule    NYC  1920  only  -      Mar   lastSun 2:00    1:00   D
Rule    NYC  1920  only  -      Oct   lastSun 2:00    0       S
Rule    NYC  1921  1966  -      Apr   lastSun 2:00    1:00   D
Rule    NYC  1921  1954  -      Sep   lastSun 2:00    0       S
Rule    NYC  1955  1966  -      Oct   lastSun 2:00    0       S
# Zone  NAME                GMT+00  RULES  FORMAT  [UNTIL]
Zone America/New_York      -4:56:02 -    LMT    1883 November 18, 12:03:58
                        -5:00    US      E%sT    1920
                        -5:00    NYC      E%sT    1942
                        -5:00    US      E%sT    1946
                        -5:00    NYC      E%sT    1967
                        -5:00    US      E%sT

```

Data stored for each zone

For each timezone that has multiple offsets (usually due to daylight saving time), the tz database records the exact moment of transition. The format can accommodate changes in the dates and times of transitions as well. Zones may have historical rule changes going back many decades (as shown in the example above).

Zone.tab

The file `zone.tab` is in the public domain and lists the zones. Columns and row sorting are described in the comments of the file, as follows:

```

# This file contains a table with the following columns:
# 1. ISO 3166 2-character country code. See the file `iso3166.tab'.
# 2. Latitude and longitude of the zone's principal location
#    in ISO 6709 sign-degrees-minutes-seconds format,
#    either +-DDMM+-DDDMM or +-DDMMSS+-DDDMMSS,
#    first latitude (+ is north), then longitude (+ is east).
# 3. Zone name used in value of TZ environment variable.
# 4. Comments; present if and only if the country has multiple rows.
#
# Columns are separated by a single tab.

```

```
# The table is sorted first by country, then an order within the country that
# (1) makes some geographical sense, and
# (2) puts the most populous zones first, where that does not contradict (1).
```

Data before 1970

Data before 1970 aims to be correct for the city identifying the region, but is not necessarily correct for the entire region. This is because new regions are created only as required to distinguish clocks since 1970.

For example, between 1963-10-23 and 1963-12-09 in Brazil only the states of Minas Gerais, Espírito Santo, Rio de Janeiro, and São Paulo had summer time. However, a requested split from America/Sao_Paulo was rejected in 2010 with the reasoning that, since 1970, the clocks were the same in the whole region.^[19]

Time in Germany, which is represented by Europe/Berlin, is incorrect for the year 1945 when the Trizone used daylight saving time rules different from Berlin's.^[20]

Coverage

Zones covering multiple post-1970 countries

There are two zones that cover an area that was covered by two countries after 1970. The database follows the definitions of countries as per ISO 3166-1, whose predecessor, ISO 3166, was first published in 1974.

- Asia/Aden – two countries until 1990: North Yemen (ISO 3166-1: YE; capital Sana'a) and South Yemen (People's Republic, ISO 3166-1: YD, ISO 3166-3: YDYE; capital: Aden).
- Europe/Berlin – two countries until 1990: East Germany (ISO 3166-1: DD, ISO 3166-3: DDDE) and West Germany (ISO 3166-1: DE)

Maintenance

The tz reference code and database is maintained by a group of volunteers. Arthur David Olson makes most of the changes to the tz reference code. Paul Eggert makes most of the changes to the tz database. Proposed changes are sent to the tz mailing list, which is gatewayed to the comp.time.tz Usenet newsgroup. Source files are distributed via the IANA FTP server. Typically, these files are taken by a software distributor like Debian, compiled, and then the source and binaries are packaged as part of that distribution. End users can either rely on their software distribution's update procedures, which may entail some delay, or obtain the source directly and build the binary files themselves. The IETF has published RFC 6557 (<https://www.rfc-editor.org/rfc/rfc6557>), "Procedures for Maintaining the Time Zone Database" documenting best practices based on similar principles.

Unix-like systems

The standard path for the timezone database is `/usr/share/zoneinfo/` in Linux distributions, macOS, and some other Unix-like systems.

Usage and extensions

Boundaries of timezones

Geographical boundaries in the form of coordinate sets are not part of the tz database, but boundaries are published by Evan Siroky^[1] in GeoJSON and shapefile formats.

Use in other standards

The Unicode Common Locale Data Repository (CLDR) refers to zones in the tz database. However, as the name for a zone can change from one tz database release to another, the CLDR assigns the UN/LOCODE for the city used in the name for the zone, or an internally-assigned code if there is no such city for the zone, to a tzdb zone.^{[21][22]}

Use in software systems

The tz database is used for time zone processing and conversions in many computer software systems, including:

- BSD-derived systems, including FreeBSD, NetBSD, OpenBSD, DragonFly BSD, macOS, and iOS (they also use the reference TZ database processing code as their TZ POSIX API implementation);
- the GNU C Library and systems that use it, including GNU, most Linux distributions, BeOS, Haiku, Nexenta OS, and Cygwin;
- System V Release 4-derived systems, such as Solaris and UnixWare;
- AIX 6.1 and later^{[23][24]} (earlier versions of AIX, starting with AIX 5.2, include zoneinfo,^[25] for support of third-party applications such as MySQL,^[26] but do not use it themselves^{[25][27]});
- Android^[28]
- several other Unix systems, including IRIX, Tru64, SunOS 4.x,^[29] and UNICOS/mp;
- OpenVMS;
- the Java Runtime Environment since release 1.8 (2014), see `java.time.ZoneId` (<https://docs.oracle.com/javase/8/docs/api/java/time/ZoneId.html>)
- the Perl modules `Date::TimeZone` (<https://metacpan.org/module/Date::TimeZone>) and `Date::LeapSecond` (<https://metacpan.org/module/Date::LeapSecond>) since 2003;
- PHP releases since 5.1.0 (2005);
- the Ruby Gem TZInfo (<http://rubygems.org/gems/tzinfo>);

- the Python standard library `zoneinfo` (<https://docs.python.org/3/library/zoneinfo.html>) module, the first-party `tzdata` (<https://pypi.org/project/tzdata/>) package, and the third-party `pytz` (<https://pypi.python.org/project/pytz/>) package;
- the JavaScript language specification for Internationalization explicitly specifies the usage of IANA Time Zone names for API, and recommends the usage of the time zone data as well.^[30]
 - Numerous libraries also available: `timezone-js` (<https://github.com/mde/timezone-js>), `BigEasy/TimeZone` (<https://github.com/bigeasy/timezone>), `WallTime-js` (<https://github.com/sproutsocial/walltime-js>) and `moment-timezone` (<http://momentjs.com/timezone/>);
- the Pandas (Python) module `pandas – Python Data Analysis Library` (<http://pandas.pydata.org>);
- the .NET Framework libraries `NodaTime` (<http://nodatime.org/>), `TZ4Net` (<https://web.archive.org/web/20140201215458/http://www.babiej.demon.nl/Tz4Net/main.htm>) and `zoneinfo` (<http://zoneinfo.codeplex.com/>) Archived (<https://web.archive.org/web/20171224045950/http://zoneinfo.codeplex.com/>) 24 December 2017 at the Wayback Machine;
- the Haskell libraries `timezone-series` (<http://hackage.haskell.org/package/timezone-series>) and `timezone-olson` (<http://hackage.haskell.org/package/timezone-olson>);
- the Erlang module `ezic` (<https://github.com/drfl00b/ezic>);
- The Go standard library `time` (<https://golang.org/pkg/time/>) package;
- The Rust crate `chrono-tz` (<https://crates.io/crates/chrono-tz>);
- The Squeak Smalltalk `time` (<http://wiki.squeak.org/squeak/1076>) package;
- The C++ libraries `Boost` and `Qt`, and C++20 chrono standard library's `std::chrono::tzdb`;
- The Delphi and Free Pascal library `TZDB` (<https://github.com/pavkam/tzdb>);^[31]
- The Free Pascal library `PascalTZ` (<https://github.com/dezlov/pascaltz>);
- The Tool Command Language has a `clock` command (<https://www.tcl.tk/man/tcl/TclCmd/clock.htm#M78>) using `tzdata`;
- Oracle releases since 10g (2004);^[32]
- PostgreSQL since release 8.0 (2005);
- the Microsoft SQL Server library `SQL Server Time Zone Support` (<https://github.com/mj1856/SqlServerTimeZoneSupport>);
- MongoDB since release 3.6;
- the Dart/Flutter `Timezone` package in pub (<https://pub.dev/packages/timezone>);
- embedded software such as the firmware used in IP clocks.

The Olson timezone IDs are also used by the Unicode Common Locale Data Repository (CLDR) and International Components for Unicode (ICU). For example, the CLDR Windows–Tzid table maps Microsoft Windows time zone IDs to the standard Olson names, although such a mapping cannot be perfect because the number of time zones in Windows systems is significantly lower than in the IANA TZ database.^[33]

History

The project's origins go back to 1986 or earlier.^[34]

2011 lawsuit

On 30 September 2011, a lawsuit, *Astrolabe, Inc. v. Olson et al.*, was filed concerning copyright in the database.^{[35][36]} As a result, on 6 October 2011, the database's mailing list and FTP site were shut down.^[37] The case revolved around the database maintainers' use of *The American Atlas*, by Thomas G. Shanks, and *The International Atlas*, by Thomas G. Shanks and Rique Pottenger. It complained of unauthorised reproduction of atlas data in the timezone mailing list archive and in some auxiliary link collections maintained with the database, though it did not actually point at the database itself. The complaint related only to the compilation of historical timezone data, and did not cover extant tzdata world timezone tables.^{[36][38][39]}

This lawsuit was resolved on 22 February 2012 after the involvement of the Electronic Frontier Foundation, when Astrolabe voluntarily moved to dismiss the lawsuit without having ever served the defendants and agreed to a covenant not to sue in the future.^[40]

Move to ICANN

ICANN took responsibility for the maintenance of the database on 14 October 2011.^[4] The full database and a description of plans for its maintenance are available online from IANA.^[41]

See also

- List of tz database time zones
- Time zone
- Daylight saving time

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External links

General

- *Legal time* (https://www.itu.int/dms_pub/itu-t/opb/sp/T-SP-LT.1-2015-PDF-E.pdf) (PDF), ITU, 2015.
- *The tz database home page* (<http://cs.ucla.edu/~eggert/tz/tz-link.htm>), UCLA (deprecated, see Official IANA sources below)
- *The tz mailing list archive* (<https://lists.iana.org/hyperkitty/list/tz@iana.org/>) at IANA
- tz mailing list at IANA (<https://lists.iana.org/postorius/lists/tz.iana.org/>)
- "A literary appreciation of the Olson/Zoneinfo/tz database" (<http://blog.jonudell.net/2009/10/23/a-literary-appreciation-of-the-olsonzoneinfotz-database/>) by Jon Udell

Official IANA sources

- Home page (<https://www.iana.org/time-zones>)
- FTP (<http://ftp.iana.org/tz/>)
- *rsync*, at [rsync://rsync.iana.org/tz/](https://rsync.iana.org/tz/)

Man pages

- *zic(8)* (<https://linux.die.net/man/8/zic>) – *Linux Administration and Privileged Commands Manual* (gives the syntax of source files for the tz database)
- *tzfile(5)* (<https://linux.die.net/man/5/tzfile>) – *Linux File Formats Manual* (gives the format of compiled tz database files)

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